

Section 2: Deriving the Dual

The Dual Problem

Primal Problem

$$\begin{aligned} \min_{\pi \in \mathbb{R}_+^{n \times m}} \quad & \langle \mathbf{C}, \pi \rangle \\ \text{s.t.} \quad & \pi \mathbf{1} = \mathbf{a} \\ & \pi^\top \mathbf{1} = \mathbf{b} \end{aligned}$$

Dual Problem

$$\begin{aligned} \max_{\mathbf{f} \in \mathbb{R}^n, \mathbf{g} \in \mathbb{R}^m} \quad & \mathbf{a}^\top \mathbf{f} + \mathbf{b}^\top \mathbf{g} \\ \text{s.t.} \quad & f_i + g_j \leq c_{ij}, \quad \forall i, j. \end{aligned}$$

Can also be written as: $\mathbf{f} \oplus \mathbf{g} \leq \mathbf{C}$

The vectors $\mathbf{f}, \mathbf{g}$ are called “dual potentials” (or Kantorovich, if optimal)

Symmetries:

- $\mathbf{C}$ appears in PP’s objective but DP’ constraint, vice-versa for $\mathbf{a}, \mathbf{b}$
- PP’s has $n \cdot m$ variables, $n + m$ constraints, opposite for DP